

AI-Enhanced Code ↔ Flowchart Learning System

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Project Overview

This project is an educational developer tool that converts source code into flowcharts and generates code from flowcharts. It helps students understand program logic, algorithms, and execution flow through visualization and AI-assisted explanations.

Objectives

- Convert code to flowcharts.
- Generate code from flowcharts.
- Classify programming logic using machine learning.
- Provide AI-generated explanations and learning assistance.
- Maintain user history and learning sessions.

Technology Stack

Frontend: HTML, CSS, JavaScript

Backend: Python Flask

Database: MySQL

Machine Learning: Scikit-learn, Joblib

AI Integration: Google Gemini API

Authentication: JWT

Modules

1. User Authentication
2. Dashboard and Profile Management
3. Code to Flowchart Conversion
4. Flowchart to Code Generation
5. AI Explanation Service
6. Machine Learning Logic Classification
7. History Tracking

System Architecture

User → Flask Backend → Services Layer → MySQL Database

■ Gemini AI

■ ML Model

Database Tables

- Users
- Learning Sessions
- History Records

Project Workflow

1. User logs in.
2. User submits code or flowchart.
3. Backend processes the request.
4. ML classifies logic patterns.
5. Gemini generates explanations/code.
6. Results are displayed and stored.

Advantages

- Improves programming understanding.
- Interactive learning experience.
- Visual representation of algorithms.

- AI-assisted code explanation.

Future Scope

- Multi-language support.
- Real-time collaboration.
- Voice-based explanations.
- Advanced flowchart editing.
- Cloud deployment and analytics.

Conclusion

The AI-Enhanced Code ↔ Flowchart Learning System bridges the gap between theoretical programming concepts and practical implementation through visualization, machine learning, and generative AI.